



ExpressiveTalk: Emotionally and Stylistically Enhanced Talking Face Generation

Ahmed Shamsia

Corentin Nicodème

Jesper Nyman

Mathis Patigny

Mohamed Mohamed

Serden-Yilmaz Kose

Thiranja Jayasundara

GitHub Repository

Abstract

Several systems have been introduced over the years to create a speaking face from a picture or portrait, while providing audio input for a short video. However, these systems were designed to preserve the emotion or the speaker's style. For example, *StyleTalk* ([8]) was created to capture and transfer the speaker's style, including tempo, rhythm, and pace, using any reference video and, in turn, animating a single portrait to speak in that style with a separate audio track. Such a system could not directly control the speaker's emotions. On the other hand, *EmoGen* ([6]) allowed adding emotions from a small emotion set, such as happiness, anger, and sadness, to the reference portrait and an audio clip. The system struggled to preserve the speaker's subtle speaking style while displaying unrealistic speaking faces, which is not typical in natural conversations. In Summary, *StyleTalk* provided a speaking style with no controlled emotions, while *EmoGen* provided controlled emotions without a speaking style. Therefore, *ExpressiveTalk* is proposed as a hybrid system to bridge the gap by creating a talking-face system that provides a more realistic speaking face by combining the speaker's emotions and style into a single system.

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1 Introduction

In the modern market, video content is among the most powerful tools for marketing and advertising. Businesses around the world rely heavily on video marketing to advertise their products. This is mainly effective due to the accessibility of the internet through home computers and mobile devices. Although a business may need to produce advertising and marketing videos, it may not have the resources to hire a production team and skilled actors to deliver the desired feel and emotion.

We propose *ExpressiveTalk*, an accessible platform for businesses and non-commercial users alike to manipulate the emotions and style of video subjects. The system prompts for video and audio files, then asks what emotion and style they would like to be introduced to. Moreover, the user can specify the intensity to make the difference more or less subtle. For example, a user can instruct *ExpressiveTalk* to display the subject’s emotion of surprise at 50% intensity, then manipulate the subject’s speaking style to convey a happy tone.

ExpressiveTalk is built primarily on two existing frameworks: *EmoGen* and *StyleTalk*. Their role in the project will be explained in further detail in a later section of this paper. In brief, *EmoGen* focuses on the subject’s physical facial expressions, while *StyleTalk* manipulates their speaking style. Since the scope of *ExpressiveTalk* requires both of these features, they have been fused to mitigate their limitations and produce a more comprehensive system.

As a product of this work, *ExpressiveTalk* will serve as a platform for users to manipulate the emotion and style of subjects found in face-talking videos. This is an advancement in the field of affective computing, due to its pioneering work in bridging emotion and style control while leaving the user in control of intensity.

2 Motivation

2.1 Commerical Motive

The motivation behind this project stems mainly from the impact of video-centered advertising in the modern market. Marketers rely heavily on video advertising to promote their products, as most consumers are exposed to ads on their devices. A study by researchers at the Jyväskylä University of Applied Sciences (JAMK) found that 59% of businesses use video marketing [1]. However, among those who do implement video marketing, only 64.2% of them produce their marketing videos in-house.

ExpressiveTalk provides a platform for businesses to modify their advertising and marketing videos to make them appear more endearing to their audience. For example, a car dealership could modify its marketing videos to appear more enthusiastic and friendly, while a law firm could modify its videos to appear more serious and mature.

2.2 Personal Motive

Personal utility can also be achieved with *ExpressiveTalk*; the use cases are not limited to commercial purposes. *ExpressiveTalk* can modify the faces of a photograph’s subjects, making them display a different emotion. For example, it can make party participants appear happier or more surprised. It can also take family photographs to make the subjects display emotions at a user-defined level of intensity.

3 Research Plan

Over the past few years, a number of talking-face systems have emerged that can turn a single image or portrait into speech using a short audio clip. Most of these systems were designed with one clear priority in mind: either to preserve the original speaking style or to preserve the original emotion. For instance, *StyleTalk* [8] focuses on capturing how a person speaks, tempo, rhythm, and overall pacing, from a reference talking video, then making a still portrait speak in that same style while using a different audio track. In other words, it is very good at style transfer, but it does not let the user say “make this person look happy” or “add anger” in a direct, controllable way. Emotion is implicit, not explicit.

On the other hand, *EmoGen* [6] was built to allow the user to inject an emotion (e.g., happy, angry, sad) into a portrait along with speech. This restores emotional control, but at a cost: the system does not always preserve the speaker’s original nuances, and the generated faces can look a bit off or exaggerated compared to how people actually talk. In short, *StyleTalk* gives you style without controllable emotion, and *EmoGen* gives you emotion without reliable style. This leaves a gap between “natural but not expressive” and “expressive but not natural.”

To address this gap, we propose *ExpressiveTalk*, a hybrid talking-face framework that combines the strengths of both lines of work. The core idea is simple: let the user control *what* emotion to express and keep *how* the person speaks.

ExpressiveTalk is implemented as a web-based application. The user provides a single portrait as input, along with an audio clip and either a chosen emotion label or a short style reference video. The system then fuses these inputs using a lightweight emotion encoder and a style-aware transformer, supported by a lip-sync backbone for precise mouth movements. This design enables the rendering of emotions such as happiness, sadness, or subtle blends, while still capturing the tempo and intensity of speech from a reference clip, when available. By combining the controllability of *EmoGen* with the naturalism of *StyleTalk*, *ExpressiveTalk* aims to reduce the “uncanny valley” effect, in which emotionally enhanced faces can appear unnatural, producing consistent, emotionally rich talking heads.

4 Related Works

The domain of manipulating emotions and feelings of videos and audio is an increasingly popular and demanding market. Moreover, with recent advancements in Artificial Intelligence, affective computing has the opportunity to expand and deliver more realistic results more quickly. *EmoGen* [6] and *StyleTalk* [8] are the two main frameworks used to build *ExpressiveTalk*.

4.1 EmoGen

EmoGen is a pioneering framework in affective computing that enables users to modify and manipulate a subject’s emotions. It will first apply lip-synchronization to the talking faces. Then, it will use a talking face generation framework to generate videos with user-defined emotions and expressions. Although *EmoGen* is not the first work to implement this, it made considerable advancements in the field. It utilizes the six primary emotions: happiness, sadness, fear, anger, disgust, and neutral. *EmoGen* can be used on various languages; it is not limited to English speech.

4.2 StyleTalk

StyleTalk is a framework designed to manipulate the style found in a subject’s voice. Unlike emotion, style cannot be seen physically. Instead, it is heard in the voice of a subject. *StyleTalk* introduced a method for generating speaking style for existing face-to-face videos and audio. The authors of the project have developed a style encoder to extract facial motion patterns, a decoder to synthesize facial movements from speech content, and a style code. The style code is generated by a style-aware transformer that uses feed-forward layer weights. In short, *StyleTalk* can take video and audio input and manipulate the subject to speak with a user-defined style.

4.3 Combining Existing Works

ExpressiveTalk aims to reach a final product that utilizes the best of *EmoGen* and *ExpressiveTalk*. *EmoGen* offers lip-synchronization and facial emotional manipulation. On the other hand, *ExpressiveTalk* offers an auditory and vocal style based on a user-specified input. Fusing these two frameworks would enable progress in the field of affective computing.

5 Implementation

5.1 Wav2lip

5.1.1 Main idea and objectives

Wav2Lip [9] is a deep learning model designed to synchronize the lip movements of a face in a video with a given audio signal, even if the person has never been seen by the model before. It outperforms previous approaches, which were often limited to speakers present in the training data.

Previous models failed to generalize to new faces or videos ‘in the wild’ (uncontrolled). The authors identify two major causes:

- Pixel-level loss functions do not effectively penalize audio-visual misalignment.
- Synchronization discriminators were weak (e.g., only 56% accuracy for LipGAN [14]).

Wav2Lip solves this by introducing a ‘lip-sync expert discriminator’, a pre-trained network (from SyncNet[3]) specialized in detecting audio/video desynchronization, achieving 91% accuracy [14].

5.1.2 Model architecture

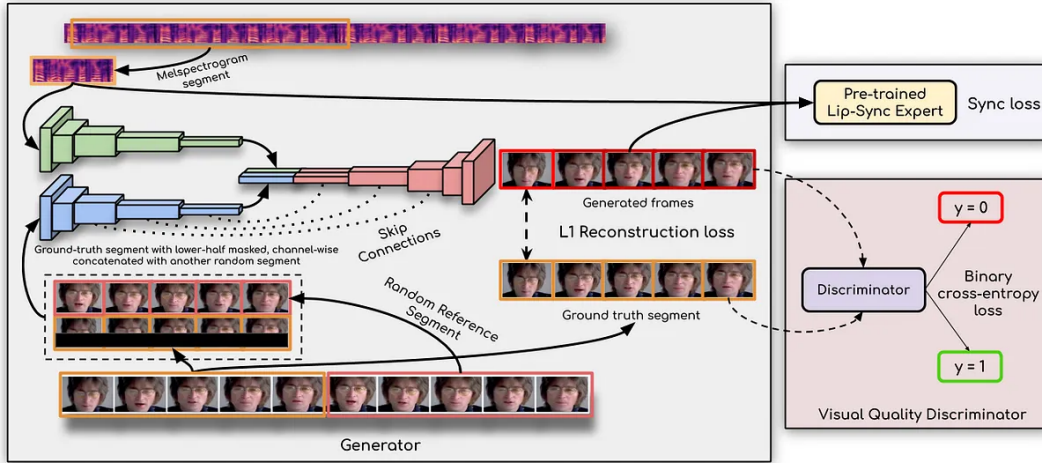


Figure 1: The architecture of *Wav2Lip*. The audio branch (green) and the video branch (blue) encode their inputs into a shared latent representation, and the decoder (red) synthesizes the final talking-face frames.

The model (Fig. 1) is a neural network architecture consisting of two encoders and one decoder:

1. **Identity Encoder:** The identity encoder receives video frames of the face and a reference segment from the same speaker. It learns visual characteristics specific to the face to preserve the speaker’s identity.
2. **Speech Encoder:** The speech encoder transforms the input audio into mel-spectrograms (5.1.3) and encodes the sound sequence into vectors representing the dynamics of phonemes.
3. **Face Decoder:** The face decoder combines audio and visual embeddings and generates frames with lip movements synchronized with the sound. It uses skip connections between the identity encoder and the decoder to preserve visual details.

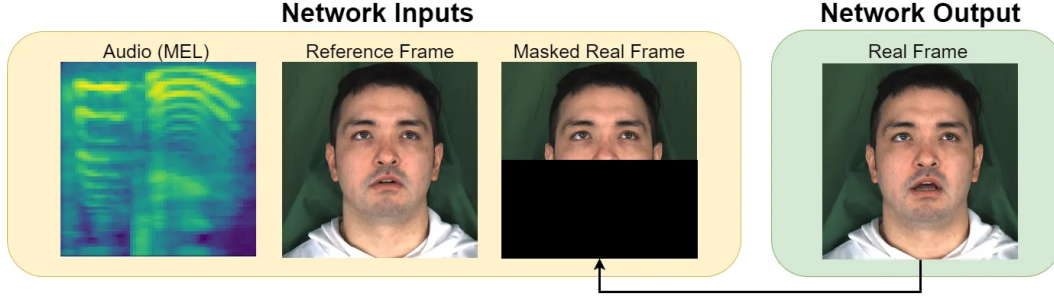


Figure 2: The inputs and output of the *Wav2Lip* generator network for a single frame.

To train the model, the lower part of the frame corresponding to the audio is masked. This means that the visual information about the subject is lost, which is why it is essential to also have another frame as a reference, taken at random in the video (there is no correspondence with the audio used as input). Thus, the encoder, which is a generator, aims to regenerate the masked part, including the mouth, so that it matches the lip movements associated with the audio.

It is also worth noting the presence of skip connections between the encoder and decoder layers (Fig. 1), which emphasize the use of the reference frame in order to facilitate realistic subject generation.

5.1.3 Mel spectrogram

A *mel* spectrogram (Fig. 3) is a visual representation of an audio signal's frequency content over time, similar to a traditional spectrogram but with a modified frequency axis based on the **mel scale** (for melody) [17] (Fig. 4). This scale is logarithmic and better approximates how humans perceive pitch, making *mel* spectrograms more suitable for tasks like speech recognition.

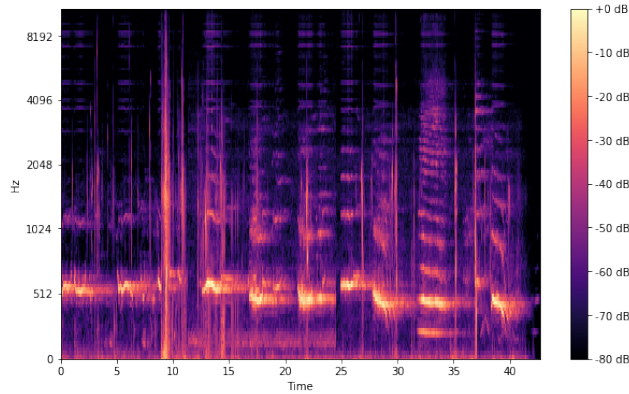


Figure 3: Example of a *mel* spectrogram of an audio [5]

In fact, the human perception of pitch is not linear: a step from 100 Hz to 200 Hz is perceived as a larger pitch change than a step from 1000 Hz to 1100 Hz, but on the *mel* scale, these perceived differences are represented as more equal.

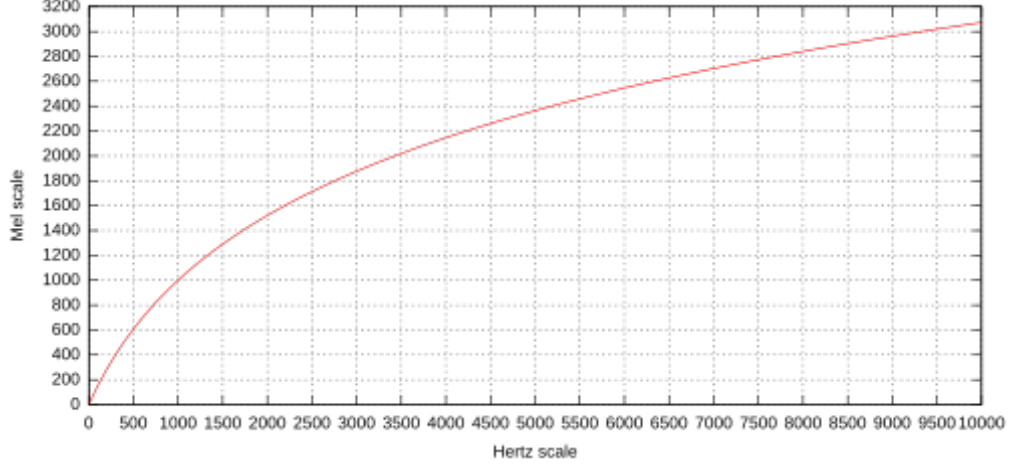


Figure 4: Plots of pitch *mel* scale versus hertz scale

The conversion formula between Hertz (Hz) and *mel* was defined by Douglas O’Shaughnessy [10]:

$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

where f is the frequency in Hertz (Hz) and m is the converted frequency in *mel*.

5.1.4 Loss functions

The generator (Fig. 1) is trained with three simultaneous objectives:

1. **Reconstruction Loss** (L1): Forces the generated frames to resemble the actual frames.
2. **Sync Loss** (with Lip-Sync Discriminator): Evaluates audio-visual consistency (cosine similarity between video and audio embeddings).
3. **Adversarial Loss** (with Visual Quality Discriminator): Improves the visual realism of the generated images.

The **generator** minimizes the weighted sum:

$$L_{\text{total}} = L_{\text{reconstruction}} + s_w \cdot L_{\text{sync}} + s_g \cdot L_{\text{adversarial}}$$

where:

- $L_{\text{reconstruction}}$ = L1 Loss (pixel-to-pixel visual fidelity)
- L_{sync} = Lip-Sync Loss (audio-visual consistency)
- $L_{\text{adversarial}}$ = Visual Quality Loss (from the visual discriminator)
- s_w, s_g = weight of each component

These three terms guarantee both visual quality and precise synchronization.

5.2 Emotions procedure

This section discusses the development and pipeline of emotion transfer in *ExpressiveTalk*. The *Emotion* module takes a face in an image or video and gently pushes key facial landmarks to make the face read as a target emotion (e.g., happy, sad, angry). It does not redraw the face; instead, it computes small, anatomically sensible displacements (corners of the mouth, eyebrows, eyelids, jaw, cheeks) and then warps the original pixels accordingly. The result feels natural because the texture, lighting, and identity stay intact, while the geometry conveys the chosen expression.

5.2.1 Pipeline

The pipeline has three stages:

1. **Landmark detection:** We use MediaPipe FaceMesh in streaming mode with one face per frame and refined landmarks enabled. To keep the model responsive, frames wider than 1280 px are down-scaled before inference. The detector returns 468 2D landmarks; we store them in pixel coordinates for direct warping. If a frame momentarily fails to detect a face (due to motion blur or occlusion), we fall back to the last valid landmarks, which prevents distracting “pops” in the video.
2. **Expression displacement:** Each target emotion is encoded as a lightweight, hand-crafted profile: a function that maps the current landmarks to a displacement field. Profiles only modify the areas that matter for expression (mouth, brows, eyelids, jawline, cheeks). To make edits scale naturally across faces and camera distances, the interpupillary distance normalizes all magnitudes; this acts as a simple, robust ruler. A global strength parameter (s) lets users dial the effect up or down without breaking the proportions.
3. **Piecewise warping & blending:** We triangulate a curated subset of control points (oval, brows, eyes, nose, inner/outer lips, jaw, cheeks) with Delaunay, then apply an affine warp per triangle from source to target. This classic piecewise-affine strategy preserves local rigidity and avoids rubbery artifacts. Finally, we feather the warped face back into the original frame using a soft convex-hull mask (a strong Gaussian blur), so edges disappear and the result looks like it was never edited.

5.2.2 Emotion profiles at a glance

Each profile is intentionally minimal and intuitive:

- **Neutral:** zero displacement (useful as a baseline or to fade effects).
- **Happy:** lift mouth corners, add gentle cheek raises, narrow lower eyelids a touch.
- **Sad:** pull brows up and in, lower the mouth center slightly, relax cheeks.
- **Fearful:** lift and angle brows, pull mouth sideways a tiny bit, open eyes more than usual.
- **Angry:** push brows down and toward the center, tighten lower eyelids, widen the nasal base.
- **Disgusted:** lift upper lip, flare the nasal area, add asymmetry cues in cheeks/brows.
- **Surprised:** raise brows and upper eyelids, drop the jaw, and open the mouth vertically.

All of the above are small geometric shifts that cue the intended effect without overfitting to any one subject. As shown in Fig. 5, an example of our emotion transfer on a single identity illustrates neutral through six basic emotions. In the figure, the system keeps texture and lighting intact and suggests the target affect by small, landmark-driven geometric edits with piecewise-affine warping and soft blending.



Figure 5: Qualitative results on a single identity. From left to right: *Neutral*, *Happy*, *Sad*, *Fearful*, *Angry*, *Disgusted*, and *Surprised*.

5.2.3 Practical details and choices

Single-face assumption: For reliability and speed, we track one face per frame. Extending to multi-face is straightforward: run detection per face and warp each region independently.

Robustness: Two small decisions help stability in the wild: (1) down-scale very large frames before detection to keep landmark noise low and runtime steady; (2) cache the last valid landmarks and reuse them if one frame drops, which makes the edit look temporally coherent.

Speed/quality trade-offs: Piecewise affine warping with Delaunay is both fast and predictable. It preserves the original skin texture (no resynthesis), keeping results realistic even on lower-resolution videos.

User controls: In addition to the target emotion, the user can set a global strength and, optionally, an output FPS when processing videos. Strength scales the displacement field linearly; in practice, values around 0.7–1.0 read clearly without looking exaggerated.

5.2.4 Video processing and visualization

The video process method reads frames, applies the full pipeline, and writes an MP4 at the source resolution with either the input FPS or a user-specified one. For quick qualitative checks, it renders a gallery (neutral and six basic emotions) for a single input image, allowing one to compare profiles side by side.

Two principles make this approach feel natural: (1) normalize all edits by a face-intrinsic distance (pupil-to-pupil) so the same numbers behave sensibly on different people and lenses; (2) move geometry, keep texture. By warping the original pixels instead of generating new ones, lighting, skin detail, and identity remain consistent, which the eye reads as “real”.

5.2.5 Limitations and extensions

The system edits 2D geometry only: it does not relight the face or re-shade teeth/inner mouth when the jaw opens, so extreme changes can reveal texture stretching. It also assumes a reasonably frontal view; heavy profile poses or large occlusions will degrade landmark quality. Natural next steps include multi-face support, temporal smoothing of displacements, and lightweight photometric adjustments (e.g., slight shadowing under a raised brow) for extra depth.

5.3 Style pipeline

The current *StyleTalk* pipeline takes as input a style video, audio, and a reference video. Described here are the steps of the pipeline:

1. **Phoneme extraction:** The first step of the pipeline is phoneme extraction. *StyleTalk* requires a JSON file of the phoneme sequence that matches the input audio. In addition, the phonemes need to be aligned to the frames of the video. For this purpose, we utilize the Bournemouth Forced Aligner [2]. This tool outputs the phonemes and their timestamps in the audio, which we then use to align the phonemes to the frames of the video. The tool also takes as input the text transcription of the audio, for which we use Whisper from OpenAI. The phonemes output by Bournemouth Forced Aligner are based on the International Phonetic Alphabet (IPA). For *StyleTalk*, these phonemes are first mapped to ARPABET phonemes and then to integer IDs using *phindex.json* found in the *StyleTalk* GitHub repository. These IDs are then saved to a json file.
2. **3DMM parameter extraction:** In this step we obtain the 3D Morphable Model (3DMM) coefficients that describe facial geometry and pose in the style video. For this purpose we use the *Deep3DFaceRecon* tool, based on a paper by Deng et al. [4]. The first step in this process is detecting facial landmarks in each frame of the video. For this purpose we use a script (*extract_kp_videos.py*) originally from the repository of the PIRender tool [12], but heavily modified for this project’s purposes. This information is then used by another script (*face_recon_videos.py*) to estimate the 3DMM coefficients, which are output into a .mat file, which can be used by *StyleTalk*. This script also takes a face from the reference video and saves it as a png file to be used by *StyleTalk*.
3. **Run StyleTalk:** In the final step, the generated phoneme sequence, style parameters and reference face image are used to generate a talking face video that matches the target identity and speaking style. Using the *inference_for_demo.py* script from *StyleTalk*, the trained model is loaded and the video generated.

5.4 Integration

The integration was done in two steps. The first step served as the central coordination module, uniting emotion transformation and lip synchronization into a cohesive pipeline for expressive talking-face generation. The design followed a modular approach, with separate functions for audio pre-processing, visual warping, emotion application, and lip-sync reference, all linked in a single workflow.

The audio pre-processing component began by loading the input waveform and converting it to a *mel* spectrogram, capturing perceptually relevant acoustic features. A pre-emphasis filter was applied to the waveform to emphasize high-frequency components that improved the speech clarity, which was mathematically described as:

$$y[t] = x[t] - \alpha x[t - 1]$$

where $\alpha = 0.97$ [11]. The script then computed the Short-Time Fourier Transformation (STFT) and mapped the resulting spectrum to the *mel* scale [16].

On the visual side, the piecewise affine warping function handled geometric transformations of facial regions. Using Delaunay triangulation, it divided the face into small triangles and computed affine transformation matrices to warp each triangle from source to destination coordinates. This ensured natural deformation of non-rigid facial regions, such as the mouth and the cheeks, preserving their realism [15]. Mathematically, this could be introduced as follows:

$$\begin{bmatrix} x_t \\ y_t \end{bmatrix} = A \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

Here, A is the affine transformation matrix estimated for each triangle. The stage of emotion transfer built and manipulated the facial landmarks to express emotions such as happy, sad, angry, or surprised [7]. When activated, it adjusted landmark positions frame by frame before the lip synchronization took place. Although the feature worked, the resulting emotional expression remained subtle; hence, the second part of the integration was completed.

The *Wav2Lip* inference function performed the core lip synchronization. It loaded a pretrained *Wav2Lip* model, detected faces in the input, aligned mel-spectrogram features with corresponding frames, and synthesized mouth movements conditioned on speech features. The model is based on a Generative Adversarial Network (GAN) framework that learns to produce realistic, temporally consistent lip movements [13]. The system then merged the generated frames with the input audio track using FFmpeg, producing a fully synchronized output.

The second stage integrated the *StyleTalk* framework with the *Wav2Lip* inference module to produce expressive, stylistically enhanced talking face videos. In this phase, the output from the *StyleTalk* repository, which included stylized facial frames generated through neural style transformation, was combined with lip-synchronized frames generated by *Wav2Lip* [13].

The integration workflow began after *Wav2Lip* completed the inference. The model first produced a temporally aligned sequence of video frames that animated the mouth region using mel-spectrogram features extracted from the input audio. Simultaneously, the *StyleTalk* model processed the same input, framed as the intermediate video output, to apply stylistic and texture-based transformations. Once both sets, the lip-synced and the stylized, were generated, the integration script extracted corresponding frame sequences and performed pixel-wise, frame-level blending. Blending ensured that stylistic cues, such as shading, tone, and facial expressivity, were seamlessly merged with the precise mouth articulation from *Wav2Lip*.

Mathematically, this blending operation is expressed as follows.

$$I_{final} + \beta I_{style} + (1 - \beta) I_{lipsync}$$

Where I_{final} is the resulting fused frame, I_{style} is the output from *StyleTalk*, $I_{lipsync}$ is the corresponding frame from *Wav2Lip*, and $\beta \in [0, 1]$ controls the degree of stylistic influence.

Through the process, the integration fuses expressive style and precise articulation, bridging the gap between realistic lip motion and personalized emotional aesthetics. The result is more engaging, lifelike talking-face synthesis, where users can not only synchronize speech but also adjust the visual tone or emotional depth of the output.

5.5 Website

For this project, we chose to showcase it on a website as it would be accessible and very easy to use that way. After reviewing different libraries and website setup options, we finally decided to use *Streamlit*. It is an easy-to-use library that lets you build a website from a Python-coded *GitHub* repository. This made it perfect for us, as we wouldn't really need to learn or explore different programming languages to set up the web application, and we were already planning to use *GitHub* for this project. Furthermore, they also host the website for free as long as it's not too big.

The website can be accessed using this link: <https://expressivetalk.streamlit.app/>

To use it, you first have to wake up the app if it is asleep, as *Streamlit* puts apps to sleep when they haven't been used for a while to reduce energy consumption.

When this is done, or if the webpage is not asleep, you have to tick the boxes, upload the required files, and choose the parameters. Then you press the "process and play" button, and after creating your video, it will display on the webpage. You can even download it if you want.

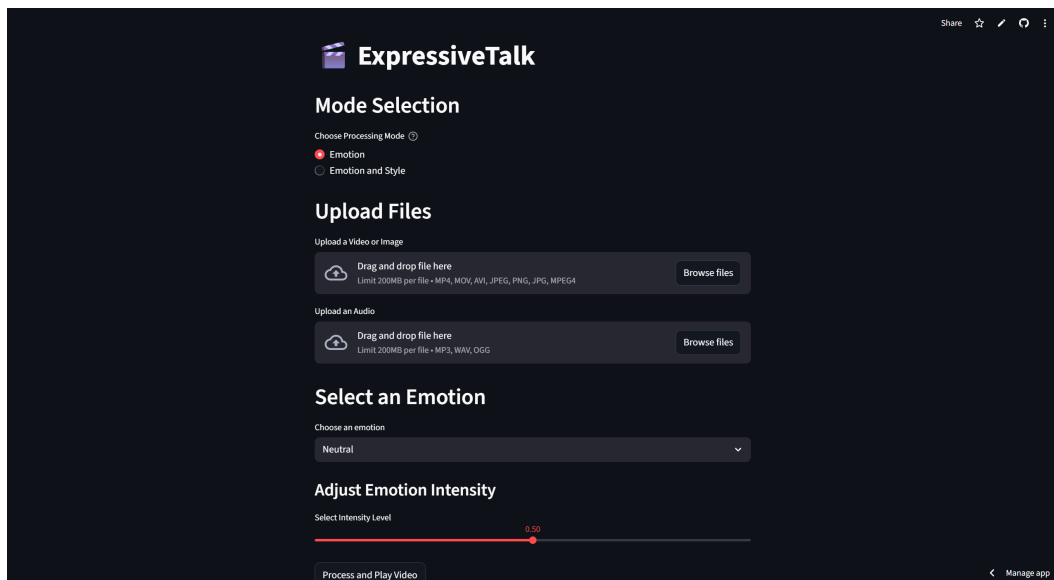


Figure 6: Screenshot of the webpage

There are obviously some issues that come with the solution we used to host the webpage. Some of the main issues with this approach are as follows:

- The time it takes to generate the output video can be quite long, since the program runs on the server and doesn't have outstanding specs.
- There is a limit on the amount of data storage allowed.
- It can be a bit annoying to test how the website is working when updating it, since you have to push the changes to *GitHub* to see them on the website.

6 Evaluation and Limitations

ExpressiveTalk aimed to provide a system that enables users to manipulate the emotion and speaking style conveyed by the subjects of face-talking videos. This section of the paper will elaborate on the performance, strengths, and limitations found in *ExpressiveTalk*.

6.1 Evaluation

The following subsections evaluate *ExpressiveTalk* in multiple metrics; the different successful tasks completed by *ExpressiveTalk* will be discussed.

6.1.1 Emotion Manipulation

ExpressiveTalk utilized the *EmoGen* framework to alter the subject’s facial emotion. In several tests using different videos, audio, emotions, and intensities, the output has been consistent with the inputs. To begin with, the lip-synchronization appears to work very well; the audio is well aligned with the subject’s lips and face, making it appear mostly realistic and natural. Moreover, the emotional manipulation was also visible. The subject’s face changes in the output frame, reflecting the user-defined emotion input. Finally, the utility of the intensity input was also visible. Setting the emotion to "anger" and the intensity to 50% would yield different results than setting the emotion to "anger" and the intensity to 100%. In the latter case, the subject would have their eyebrows lowered and stressed more, which is an indication and display of anger. This is how *ExpressiveTalk* was able to manipulate the emotions of a subject in a face-talking video.

6.1.2 Style Manipulation

The other integral part of *ExpressiveTalk*’s functionality is the manipulation of speaking style. When compared to emotion, speaking style cannot be seen visually. Instead, it is heard audibly, as speaking style consists of tone, pitch, cadence, and other features. The style manipulation was less apparent in the conducted tests. Although there is certainly a difference between styles, distinguishing the deviation from the original audio is harder because the differences are more subtle. Moreover, audible changes are generally less noticeable than visible modifications. For example, the same input video and audio were submitted with two speaking styles: happy and angry. In the former, the speaker has a higher pitch. However, the former will have a lower pitch and a more aggressive tone. This indicates that *ExpressiveTalk* was able to deliver manual style manipulation outside the web application.

6.2 Limitations

The following subsections discuss the limitations found in *ExpressiveTalk*, while respecting the potential scope of such a project.

6.2.1 StyleTalk

Due to some limitations, *StyleTalk* functionality was not included in the demo website. The limitations stemmed from several factors that made integrating *StyleTalk* very difficult. For example, *StyleTalk*, *Deep3DFaceRecon*, and the other tools had various old and partially conflicting dependencies. The tools also used outdated Python modules with features that no longer work. This had to be fixed by manually patching some obsolete lines of code in *Deep3DFaceRecon_pytorch*, for instance. Multiple pre-trained models and other assets also had to be downloaded, and the Basel Face Model used by *Deep3DFaceRecon_pytorch* required the user to request access. There is a script (*styletalk/setup_environment.sh*) in place to automate fetching all dependencies and assets, but due to time constraints, the integration work was not fully completed. This means that while the *StyleTalk* pipeline is functional, it has to be run manually.

6.2.2 Multi Subject Videos

ExpressiveTalk was built on the basis that only one subject would be present in a video. However, there are valid reasons a user may want to use a video featuring two different subjects. *ExpressiveTalk* cannot manipulate multiple subjects in a single video.

6.2.3 Lack of Application Programming Interface

An application programming interface (API) is a set of rules that allows different applications to share data. APIs can be especially useful when data is being shared remotely, across two computers in various locations. *ExpressiveTalk* offers only a graphical, web-based platform. In other words, a user would not be able to make an API call to seamlessly upload their content, and then receive the results in a software, likely developed and used by the user. Creating an API for *ExpressiveTalk* would make it a more appealing and inviting tool by expanding its utility.

7 Discussion

ExpressiveTalk was intended to fuse emotion manipulation and speaking style in talking-face videos. The system successfully modified facial expressions in accordance with the emotions and intensities specified by the user. Different inputs yield different outputs; a user submitting "happy" and "angry" on the same video will have visibly different results. Moreover, the platform was intended to manipulate the speaking subject's speaking style in pitch, tone, and cadence.

However, the manipulation of speaking style was not fully integrated in the *ExpressiveTalk* system. Attempts were made to incorporate *StyleTalk*, an existing framework for managing spoken styles, into the new system. Yet, it has not been fully integrated, and can only be run manually outside of *ExpressiveTalk*. Thus, users cannot benefit from the seamless style modification in the web-based application.

The system also constrains the user to single-subject videos. In other words, *ExpressiveTalk* cannot be applied to videos with multiple subjects, and emotions or speaking styles of different subjects cannot be controlled. This may significantly reduce the platform's potential, as many users may want to use videos with multiple subjects. Moreover, the platform lacks an application programming interface (API), limiting the system's utility and extensibility. Currently, it can only be used in the graphical web-based application, which can be slower and less convenient if a user would like to use *ExpressiveTalk* for a large number of videos.

Despite these limitations, *ExpressiveTalk* has advanced the field of affective computing by providing a platform for controlling emotion and for research on manipulating speaking style. Although the latter has not been fully integrated into the web application, progress has been made to improve the existing frameworks. The new system attempted to mitigate the limitations of existing frameworks by making the work mutual.

8 Conclusion

This report presents *ExpressiveTalk*, a new system that finally connects emotional control and style manipulation in talking-face videos. Unlike existing frameworks, such as *EmoGen* and *StyleTalk*, *ExpressiveTalk* merged the features of both frameworks to mitigate their limitations. By combining *EmoGen*'s emotional control with *StyleTalk*'s style manipulation, the new system enables users to create more expressive, natural videos tailored to their specifications. While full integration was not achieved, the work demonstrated the possibility of integrating emotional control with speaking style manipulation.

The motive behind such a system lies in the rise of video marketing and advertising. Most businesses rely on videos to promote their products; thus, their videos must be natural-looking and convey the proper emotion and speaking style. Moreover, *ExpressiveTalk* can also be used for personal purposes. It can be used to modify and manipulate family photos, making a subject display a particular emotion. This is a more inviting feature, since users can specify the intensity of a feeling. This platform enables the generation of a more attractive, tantalizing video by adjusting emotional and stylistic features.

ExpressiveTalk successfully controlled emotions in talking-face videos, and initial work was conducted to manipulate speaking style. The emotional element of the system is evident when comparing results with different user-set emotions. When set to happy, the face will display a smile; when set to surprise, it will display wider eyes; and when set to anger, it will display lowered eyebrows. Moreover, when the emotion remains the same but the intensity is reduced, the new result will fall between the original video and the high-intensity version. The stylistic element of the system also worked when listening to two different outputs. When the style is set to happy, the subject's voice

will be higher-pitched. On the other hand, if the style were set to angry, the subject’s voice would be lower-pitched. These are the different ways *ExpressiveTalk* can modify talking-face videos.

Although *ExpressiveTalk* has made advances in this field, it does have some limitations. The main limitation of the system is its restriction to single-subject videos; *ExpressiveTalk* cannot be used on videos with multiple subjects, nor can it control the emotions or manipulate the speaking style of various subjects simultaneously. Another limitation of *ExpressiveTalk* is its lack of extensibility via an application programming interface (API). Currently, *ExpressiveTalk* can only be accessed and used via a graphical web app. This reduces the ways a user can use the system. Although *ExpressiveTalk* provides a solid platform for video manipulation, it still has some lacking features.

In conclusion, *ExpressiveTalk* advances the field of affective computing by integrating emotional control and style manipulation into a single framework. Through this, users can create more natural, controllable, and extensible videos for both commercial and personal use.

9 Team work

9.1 Organization

Our working system was based on regular Zoom meetings, at least once a week, to review everyone’s progress and decide on the next steps. As the project relies on many existing implementations, it was necessary for everyone to stay informed about how each part works. In particular, the technical aspects had to be explained so that everyone could stay up to date to the main pipeline.

We collaborate on all our code on GitHub, a repository that also serves as the direct source for the website.

9.2 Division of responsibilities

Here is the general distribution of roles within the team:

- Ahmed Shamsia: Wrapping code, *StyleTalk*, hybrid version
- Corentin Nicodème: Lip synchronization (*Wav2Lip*), slideshow, oral presentation
- Jesper Nyman: *StyleTalk* pipeline and integration work, writing
- Mathis Patigny: Website
- Mohamed Mohamed: *Emotion* code, slideshow, writing
- Serden-Yilmaz Kose: Writing, oral presentation
- Thiranjaya Jayasundara: Integration of codes with the website, writing

For this report, we decided that the technical sections (the presentation of frameworks) would be written by the people who specifically worked on them. As a result, everyone took part in its drafting.

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